Lab sheet-02

Saurabh Kumar Jha
CU24250005

```
// TASK 1: Create product array
//==================================================
const products = [

   {
      id: 1,
      name: "Headphones",
      price: 2499,
      image: "images/headphones.jpg"
   },
{
      id: 2,
      name:     "Smart     Watch",
      image:
price: 3999,
"images/watch.jpg"
   },
{
      id: 3,
      name: "Bluetooth Speaker",
price: 1999,
      image: "images/speaker.jpg"
   },
{
      id: 4,
      name: "Keyboard",
price: 3499,
      image: "images/keyboard.jpg"
   },    {
id:5,
      name: "Gaming Mouse",
price: 1499,
      image: "images/mouse.jpg"
   },
{
      id: 6,
      name: "USB Charger",
price: 999,
      image: "images/charger.jpg"
   },    {
id:7,
            name: "Laptop Stand",
price: 1299,                  image:
"images/stand.jpg"
   },
{
      id: 8,
      name: "Webcam",
      price: 2299,
      image: "images/webcam.jpg"
   }
];
//TASK3:CreatecartanduselocalStorage
//==================================================
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let cart = JSON.parse(localStorage.getItem("cart")) || []; function
saveCart() {
   localStorage.setItem("cart", JSON.stringify(cart));
}

// TASK 8: Update cart item count
// ================================================
function updateCartCount() {

   const cartCount = document.getElementById("cart-count");

   if (cartCount) {

      const count = cart.reduce(function(total, item) {
         return total + item.quantity;
      }, 0);

      cartCount.textContent = count;
   }
}

// TASK 2: Dynamically display products
// ================================================
function displayProducts() {
   const productList = document.getElementById("product-list");
   if                (!productList)                return;
productList.innerHTML              =              "";
products.forEach(function(product) {          const
card       =       document.createElement("div");
card.innerHTML = `
         <img src="${product.image}" width="150">
         <h3>${product.name}</h3>
<p>₹${product.price}</p>
         <button onclick="addToCart(${product.id})">
            Add to Cart
         </button>      `;
productList.appendChild(card);
   });
}
// TASK 3: Add product to cart
// ================================================
function addToCart(id) {
   const product = products.find(function(product) {
      return product.id === id;
   });
   const existingItem = cart.find(function(item) {
      return item.id === id;
   });    if (existingItem) {
existingItem.quantity++;
   } else {       cart.push({
id: product.id,           name:
product.name,           price:
product.price,          image:
product.image,
quantity: 1
      });
   }   saveCart();
updateCartCount();
alert("Product added to cart!");
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}

// TASK 4: Display cart items
// ================================================
function displayCart() {
   const cartItems = document.getElementById("cart-items");
if (!cartItems) return;    cartItems.innerHTML = "";    if
(cart.length === 0) {
      cartItems.innerHTML = "<p>Your cart is empty.</p>";
     calculateTotal();
return;
   }
   cart.forEach(function(item) {        const div =
document.createElement("div");
div.innerHTML                    =                    `
<h3>${item.name}</h3>                  <p>Price:
₹${item.price}</p>
         <label>              Quantity:
<input                 type="number"
min="1"
value="${item.quantity}"
               onchange="changeQuantity(${item.id}, this.value)"
            >
         </label>

         <button onclick="removeFromCart(${item.id})">
            Remove
         </button>
         <hr>
      cartItems.appendChild(div);
   });
calculateTotal();
}
// TASK 5: Change quantity
// ================================================
function changeQuantity(id, quantity) {
const item = cart.find(function(item) {
      return item.id === id;
   });    if
(item) {

      item.quantity = Number(quantity);

       if (item.quantity < 1) {
          item.quantity = 1;
       }
   }    saveCart();
displayCart();
updateCartCount();
}
// TASK 6: Remove item from cart
// ================================================
function removeFromCart(id) {    cart = cart.filter(function(item) {
return item.id !== id;
   });
saveCart();
displayCart();
updateCartCoun
t();
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}
// TASK 5: Calculate grand total using reduce()
// ================================================
function calculateTotal() {
   const totalElement = document.getElementById("cart-total");

   if (!totalElement) return;

   const total = cart.reduce(function(sum, item) {

      return sum + item.price * item.quantity;

   }, 0);

   totalElement.textContent = "₹" + total; }

// TASK 7: Checkout form validation
// ================================================
function setupCheckout() {
   const form = document.getElementById("checkout-form");
if (!form) return;
   form.addEventListener("submit", function(event) {
      event.preventDefault();
      const name = document.getElementById("name").value.trim();
const address =
         document.getElementById("address").value.trim();
const pincode =
         document.getElementById("pincode").value.trim();
const phone =
         document.getElementById("phone").value.trim();
      // Clear old errors document.getElementById("name-
      error").textContent = ""; "";
document.getElementById("address-error").textContent    =
document.getElementById("pincode-error").textContent = "";
      document.getElementById("phone-error").textContent = "";
let valid = true;       // Name validation        if (name === "") {
         document.getElementById("name-error").textContent =
            "Name is required";
valid = false;
      }
      // Address validation
if (address === "") {
         document.getElementById("address-error").textContent =
            "Address is required";
valid = false;
      }
      // Pincode validation
      if (!/^\d{6}$/.test(pincode)) {

         document.getElementById("pincode-error").textContent =
            "Pincode must be exactly 6 digits";
         valid = false;
      }
      // Phone validation           if
(!/^\d{10}$/.test(phone))          {
document.getElementById("pho
ne-error").textContent =
            "Phone must be exactly 10 digits";
         valid = false;
```

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        }
        // TASK 8: Successful order
        // ==================================================
if (valid) {
            document.getElementById("confirmation").textContent =
                "Order placed successfully!";
cart = [];
            localStorage.removeItem("cart");
updateCartCount();            form.reset();
        }
    });
}
// RUN ALL FUNCTIONS
// ==================================================
document.addEventListener("DOMContentLoaded", function() {

    displayProducts();

    displayCart();

    updateCartCount();

    setupCheckout();

});
```

